Import libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import timedelta
import warnings
warnings.filterwarnings('ignore')
import mysql.connector
```

In [10]:
```
pip install mysql-connector-python
```

```
Requirement already satisfied: mysql-connector-python in c:\users\pankaj\anaconda3\lib\site-packages (9.6.0)
Note: you may need to restart the kernel to use updated packages.
```

Load The Data

In [3]:
```
# Define your connection parameters
config = {
    'user': 'root',        # e.g., 'root'
    'password': 'pankaj12345',    # The password you set during MySQL installation
    'host': 'localhost',            # Usually 'localhost' for a local setup
    'database': 'retailanalytics' # The name of the specific database you want to use
}

try:
    # Establish the connection
    cnx = mysql.connector.connect(**config)

    if cnx.is_connected():
        print("Connection established successfully!")


        # 2. Use read_sql to execute the query and return a DataFrame
        query = "select * from customer_master"
        query_2 = "select * from transactions"
        df_cust = pd.read_sql(query, cnx)
        df_trans = pd.read_sql(query_2 , cnx)

        # Now 'df' is a clean table (DataFrame)
        print(df_cust.head())
        print(df_trans.head())




except mysql.connector.Error as err:
    print(f"Error: {err}")

finally:
    # Always ensure the connection is closed
    if 'cnx' in locals() and cnx.is_connected():
        cnx.close()
        print("Connection closed.")
```

```
Connection established successfully!
  CustomerID            Name                    Email  Gender  Age     City  \
0  CUST10000  Onkar Bhargava          pkeer@yahoo.com    Male   54    Delhi
1  CUST10001     Divit Kohli        mkalita@sarin.com  Female   48  Kolkata
2  CUST10002      Kiara Behl     apteanay@hotmail.com    Male   75  Kolkata
3  CUST10003  Vaibhav Sankar  bseshadri@choudhry.info    Male   62     Pune
4  CUST10004  Shray Dâ€™Alia   bdhillon@toor-mall.com    Male   55    Delhi

  MaritalStatus  NumChildren    joindate
0      Divorced            0  2021-02-22
1       Married            0  2023-12-06
2       Widowed            2  2023-08-23
3      Divorced            2  2022-11-17
4      Divorced            0  2022-12-04
  CustomerID TransactionDate  TransactionAmount
0  CUST10771      2023-07-31            2383.07
1  CUST10100      2024-03-10             497.54
2  CUST10031      2025-02-17             536.78
3  CUST10987      2023-07-17             314.89
4  CUST10831      2024-12-15            2543.19
Connection closed.
```

In [4]:
```
df_cust.head()
```

Out[4]:

	CustomerID	Name	Email	Gender	Age	City	MaritalStatus	NumChildren	joindate
0	CUST10000	Onkar Bhargava	pkeer@yahoo.com	Male	54	Delhi	Divorced	0	2021-02-22
1	CUST10001	Divit Kohli	mkalita@sarin.com	Female	48	Kolkata	Married	0	2023-12-06
2	CUST10002	Kiara Behl	apteanay@hotmail.com	Male	75	Kolkata	Widowed	2	2023-08-23
3	CUST10003	Vaibhav Sankar	bseshadri@choudhry.info	Male	62	Pune	Divorced	2	2022-11-17
4	CUST10004	Shray Dâ€™Alia	bdhillon@toor-mall.com	Male	55	Delhi	Divorced	0	2022-12-04

In [6]:
```
df_trans.head()
```

Out[6]:

	CustomerID	TransactionDate	TransactionAmount
0	CUST10771	2023-07-31	2383.07
1	CUST10100	2024-03-10	497.54
2	CUST10031	2025-02-17	536.78
3	CUST10987	2023-07-17	314.89
4	CUST10831	2024-12-15	2543.19

Finding Missing Values

In [7]:
```
missing_values_customers=df_cust.isnull().sum()
print('Misiing values in customer master are:\n')
print(missing_values_customers)

missing_values_transaction=df_trans.isnull().sum()
print('\nMisiing values in customer transactions are:\n')
print(missing_values_transaction)
```

```
Misiing values in customer master are:

CustomerID       0
Name             0
Email            0
Gender           0
Age              0
City             0
MaritalStatus    0
NumChildren      0
joindate         0
dtype: int64

Misiing values in customer transactions are:

CustomerID           0
TransactionDate      0
TransactionAmount    0
dtype: int64
```

In [8]:
```
missing_values_transaction=df_trans.isnull().sum()
missing_values_transaction
```

Out[8]:
```
CustomerID           0
TransactionDate      0
TransactionAmount    0
dtype: int64
```

Changing Data Type As Required For Further Analysis.

Ensuring Dates In Columns Are In Datetime Format.

In [9]:
```
df_cust.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   CustomerID     1000 non-null   object
 1   Name           1000 non-null   object
 2   Email          1000 non-null   object
 3   Gender         1000 non-null   object
 4   Age            1000 non-null   int64
 5   City           1000 non-null   object
 6   MaritalStatus  1000 non-null   object
 7   NumChildren    1000 non-null   int64
 8   joindate       1000 non-null   object
dtypes: int64(2), object(7)
memory usage: 70.4+ KB
```

In [10]:
```
df_trans.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23050 entries, 0 to 23049
Data columns (total 3 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   CustomerID         23050 non-null  object
 1   TransactionDate    23050 non-null  object
 2   TransactionAmount  23050 non-null  float64
dtypes: float64(1), object(2)
memory usage: 540.4+ KB
```

In [14]:
```
#joindate & TransactionDate column is in object datatype it should be in datetime format.

#converting joindate & TransactionDate column in datetime format.

df_cust['joindate']=pd.to_datetime(df_cust['joindate'])
df_trans['TransactionDate']=pd.to_datetime(df_trans['TransactionDate'])
```

Checking Shape & Structure

In [15]:
```
print(f"Total Customer Records In The Master Data :",df_cust.shape[0] )
print(f"Total Transactions Records :",df_trans.shape[0] )
```

```
Total Customer Records In The Master Data : 1000
Total Transactions Records : 23050
```

Checking Other Details

Keep only valid transaction

We keep transactions whose CustomerID exists in the master customer data

In [16]:
```
df_trans=df_trans[df_trans["CustomerID"].isin(df_cust["CustomerID"])].copy()
df_trans.shape[0]
```

Out[16]:
```
23050
```

Checking Duplicate Entries

In [17]:
```
duplicate=df_trans[df_trans.duplicated()]
print("\nDuplicate Entries in Transactions are:\n")
print(duplicate)
duplicate_trs=df_cust[df_cust.duplicated()]
print("\nDuplicate Entries in Customer Master are:\n ")
print(duplicate_trs)
```

```
Duplicate Entries in Transactions are:

Empty DataFrame
Columns: [CustomerID, TransactionDate, TransactionAmount]
Index: []

Duplicate Entries in Customer Master are:

Empty DataFrame
Columns: [CustomerID, Name, Email, Gender, Age, City, MaritalStatus, NumChildren, joindate]
Index: []
```

RFM Calculations

Referance Date = max transaction date + 1 day

In [18]:
```
ref_date=df_trans["TransactionDate"].max() + timedelta(days=1)
ref_date
```

Out[18]:
```
Timestamp('2025-07-30 00:00:00')
```

For Each Customer :

-LastTransDate = most recent purchase Date

-Frequency = Number of rows (Purchases)

-Monetary = Sum of Transaction Amount

In [19]:
```
# Defining temporary variable of Calculation of RFM is rfm.
rfm=(df_trans.groupby("CustomerID").agg
     (LastTransDate=("TransactionDate","max"),
      Frequency = ("TransactionDate","count"),
      Monetary = ("TransactionAmount","sum")).reset_index())
rfm.head()
```

Out[19]:

	CustomerID	LastTransDate	Frequency	Monetary
0	CUST10000	2025-07-17	23	21265.49
1	CUST10001	2025-06-25	30	28654.31
2	CUST10002	2025-07-12	24	23884.03
3	CUST10003	2025-05-10	25	24206.03
4	CUST10004	2025-07-22	19	25565.30

Calculating Recency (in days) = Difference From ref_date

In [20]:
```
rfm["Recency"]=(ref_date-rfm["LastTransDate"]).dt.days
rfm.head()
```

Out[20]:

	CustomerID	LastTransDate	Frequency	Monetary	Recency
0	CUST10000	2025-07-17	23	21265.49	13
1	CUST10001	2025-06-25	30	28654.31	35
2	CUST10002	2025-07-12	24	23884.03	18
3	CUST10003	2025-05-10	25	24206.03	81
4	CUST10004	2025-07-22	19	25565.30	8

Arranging columns as Required

In [21]:
```
rfm = rfm[["CustomerID","Recency","Frequency","Monetary"]]
rfm.head()
```

Out[21]:

	CustomerID	Recency	Frequency	Monetary
0	CUST10000	13	23	21265.49
1	CUST10001	35	30	28654.31
2	CUST10002	18	24	23884.03
3	CUST10003	81	25	24206.03
4	CUST10004	8	19	25565.30

Defining the Scores R/M/F

Recency (Lower is better)

Recency<=30 days : 5

Recency<=60 days : 4

Recency<=120 days : 3

Recency<=240 days : 2

Recency>240 days : 1

In [22]:
```
r_bins = [0,30,60,120,240,float("inf")] # creating bins for range
r_labels = [5,4,3,2,1] # creating labels for range in bins (reverced because lower recency is better)

# "pd.cut" is function in pandas which cuts values one by one and creates grouping by bins and labels

# "include_lowest=True" means include the lowest in range

# "right=True" means intervals are right closed (0,30] means 30 is included in range 0 to 30
#                                and (30,60] 30 is not included in range 30 to 60 and so on,
#                                for continous range.

# ".astype(int)" returns values as numerical integer type.

rfm["R_score"] = pd.cut((rfm["Recency"]), bins=r_bins, labels=r_labels, include_lowest=True, right=True).astype(int)

rfm.head()
```

Out[22]:

	CustomerID	Recency	Frequency	Monetary	R_score
0	CUST10000	13	23	21265.49	5
1	CUST10001	35	30	28654.31	4
2	CUST10002	18	24	23884.03	5
3	CUST10003	81	25	24206.03	3
4	CUST10004	8	19	25565.30	5

Frequency (Higher is better)

Frequency <=07 : 1

Frequency <=14 : 2

Frequency <=21 : 3

Frequency <=28 : 4

Frequency >28 : 5

In [23]:
```
f_bins = [0,7,14,21,28,float("inf")] # creating bins for range
f_labels = [1,2,3,4,5]               # creating labels for range in bins (not reverced because Higher frequency is better)

# "pd.cut" is function in pandas which cuts values one by one and creates grouping by bins and labels

# "include_lowest=True" means include the lowest in range

# "right=True" means intervals are right closed (0,7] means 7 is included in range 0 to 7
```

```
#                                and (7,14] 7 is not included in range 7 to 14 and so on,
#                                for continous range.

# ".astype(int)" returns values as numerical integer type.

rfm["F_score"]=pd.cut(rfm["Frequency"],bins=f_bins,labels=f_labels,include_lowest=True,right=True).astype(int)
rfm.head()
```

Out[23]:

	CustomerID	Recency	Frequency	Monetary	R_score	F_score
0	CUST10000	13	23	21265.49	5	4
1	CUST10001	35	30	28654.31	4	5
2	CUST10002	18	24	23884.03	5	4
3	CUST10003	81	25	24206.03	3	4
4	CUST10004	8	19	25565.30	5	3

Monetary (Higher is better)

Monetary <= 10000 : 1

Monetary <= 20000 : 2

Monetary <= 30000 : 3

Monetary <= 40000 : 4

Monetary > 40000 : 5

In [24]:

```
m_bins=[0,10000,20000,30000,40000,float("inf")]    # creating bins for range
m_labels=[1,2,3,4,5]                               # creating labels for range in bins (not reverced because Higher Monetary is better)

# "pd.cut" is function in pandas which cuts values one by one and creates grouping by bins and labels

# "include_lowest=True" means include the lowest in range

# "right=True" means intervals are right closed (0,10000] means 10000 is included in range 0 to 10000
#                                and (10000,20000] 10000 is not included in range 10000 to 20000 and so on,
#                                for continous range.

# ".astype(int)" returns values as numerical integer type.
rfm["M_score"]=pd.cut(rfm["Monetary"],bins=m_bins,labels=m_labels,include_lowest=True,right=True).astype(int)
rfm.head()
```

Out[24]:

	CustomerID	Recency	Frequency	Monetary	R_score	F_score	M_score
0	CUST10000	13	23	21265.49	5	4	3
1	CUST10001	35	30	28654.31	4	5	3
2	CUST10002	18	24	23884.03	5	4	3
3	CUST10003	81	25	24206.03	3	4	3
4	CUST10004	8	19	25565.30	5	3	3

RFM SCORE (STRING)

In [25]:
```
rfm["RFM_Score"]=(rfm["R_score"].astype(str) + rfm["F_score"].astype(str) + rfm["M_score"].astype(str))
# converting into sting beacuse we want concat value nor addition value.
rfm.head()
```

Out[25]:

	CustomerID	Recency	Frequency	Monetary	R_score	F_score	M_score	RFM_Score
0	CUST10000	13	23	21265.49	5	4	3	543
1	CUST10001	35	30	28654.31	4	5	3	453
2	CUST10002	18	24	23884.03	5	4	3	543
3	CUST10003	81	25	24206.03	3	4	3	343
4	CUST10004	8	19	25565.30	5	3	3	533

Naming the segments

In [26]:
```
def segment_row(r,f,m):
    if(r>=4) and (f>=4) and (m>=4):
        return "Champions"
    elif(f>=4) and (r>=2):
        return "Loyal"
    elif(r>=4) and (2<=f<=3):
        return "Potential Loyalist"
    elif(r<=2) and (f>=3):
        return "Risk"
    elif (m>=4) and (2<=f<=3) and (r>=3):
        return "Big Spenders"
    elif (r==1) and (f<=2) and (m<=2):
        return "Lost"
    else:
        return "Others"

rfm["Segment"]=[segment_row(r,f,m) for r,f,m in zip(rfm["R_score"],rfm["F_score"],rfm["M_score"])]
# creating segment column in rfm DataFrame.
# loop r,f,m picks values from zip function defined.
# zip function in loop Gives values from defined columns from rfm one by one and gives output in variable r,f,m in loop .
rfm.head()
```

Out[26]:

	CustomerID	Recency	Frequency	Monetary	R_score	F_score	M_score	RFM_Score	Segment
0	CUST10000	13	23	21265.49	5	4	3	543	Loyal
1	CUST10001	35	30	28654.31	4	5	3	453	Loyal
2	CUST10002	18	24	23884.03	5	4	3	543	Loyal
3	CUST10003	81	25	24206.03	3	4	3	343	Loyal
4	CUST10004	8	19	25565.30	5	3	3	533	Potential Loyalist

In [27]:
```
segment_customers = rfm["Segment"].value_counts().reset_index()
segment_customers.columns=["Segment","Customer_Count"] # changing the columns header name
```

Basic Buisness Analysis

In [28]:
```
print("\n ============= SIMPLE BUSINESS ANALYSIS ===========")
#  overall business summary

total_customers = rfm["CustomerID"].nunique() #or .shape[0]
```

```python
total_revenue = rfm["Monetary"].sum()

avg_revenue_per_customer=rfm["Monetary"].mean()

print(f"\n Total customers :{total_customers}")

print(f"\n Total Revenue : Rs {total_revenue}")

print(f"\n Average Revenue per customer  : Rs {avg_revenue_per_customer:.2f}")

# customer in each segment

print(f"\n\n Numbers of Customers in each Segments :\n\n ")
print(segment_customers)

#revenue for each segment

segment_revenue=rfm.groupby("Segment")["Monetary"].sum().reset_index()

print(f"\n\nRevenue Contribution by Each Segment :\n\n")
print(segment_revenue)
```

```
============ SIMPLE BUSINESS ANALYSIS ===========

Total customers :1000

Total Revenue : Rs 23053199.66

Average Revenue per customer  : Rs 23053.20


Numbers of Customers in each Segments :

              Segment  Customer_Count
0               Loyal             542
1  Potential Loyalist             248
2           Champions              84
3              Others              77
4                Risk              47
5                Lost               2


Revenue Contribution by Each Segment :


              Segment     Monetary
0           Champions   2791699.03
1                Lost     29221.39
2               Loyal  13381926.89
3              Others   1416376.98
4  Potential Loyalist   4495435.99
5                Risk    938539.38
```

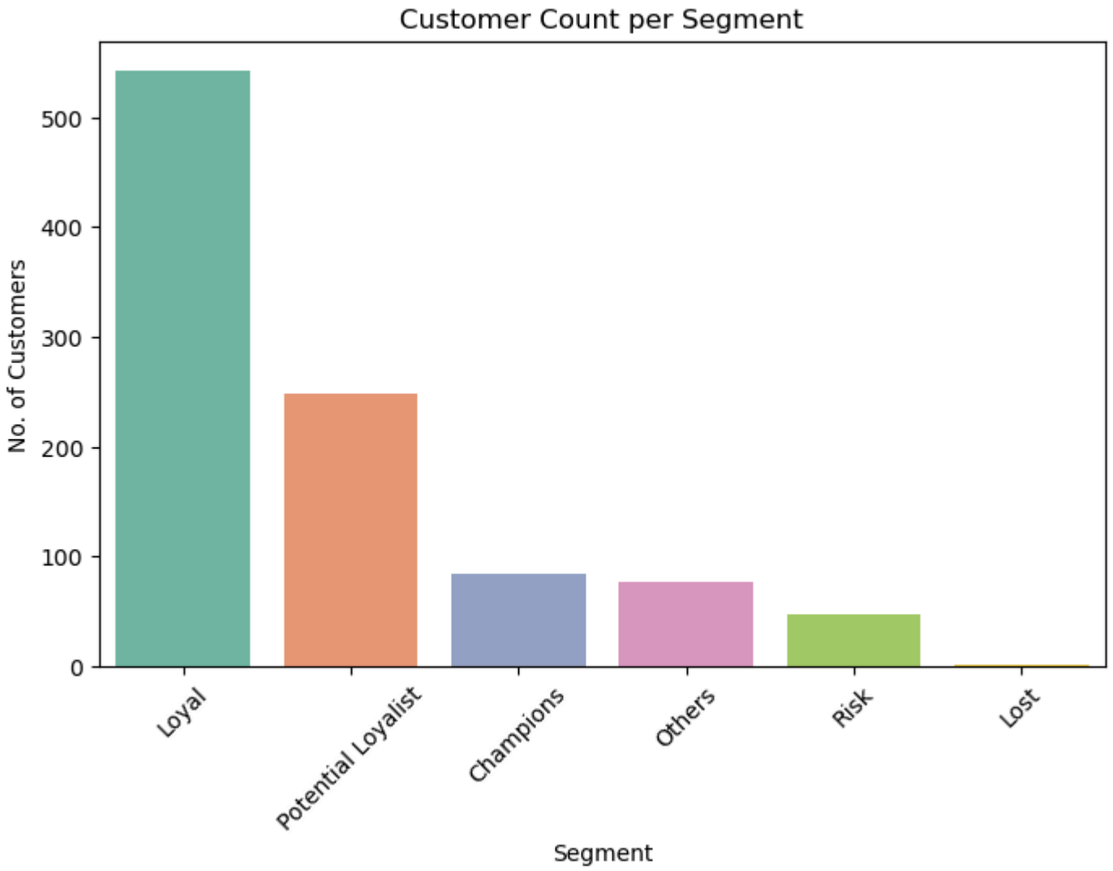
Visualisation

In [29]:

```python
# Bar chart

# printing Customer in each segment with the chart

print(f"\n\n Numbers of Customers in each Segments :\n\n ")
print(segment_customers)
```

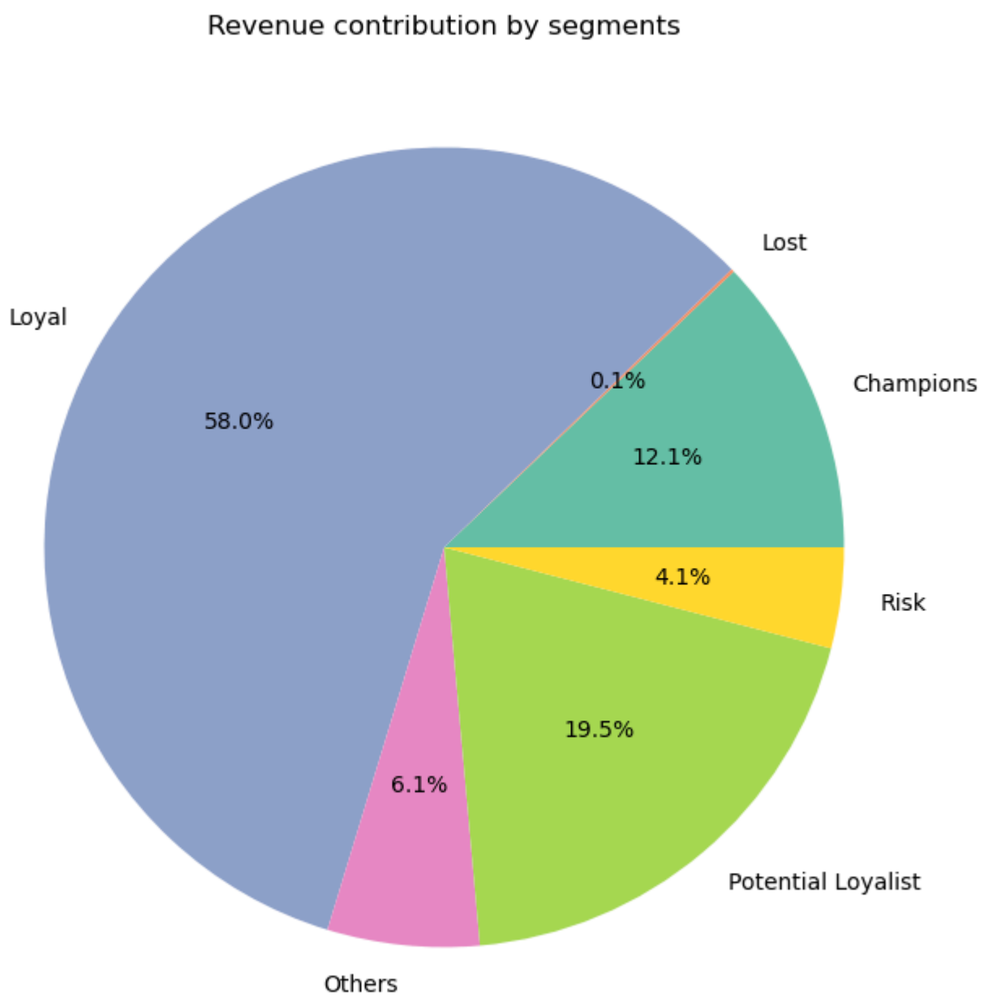
```
plt.figure(figsize=(8,5))
sns.barplot(x="Segment",y="Customer_Count",data=segment_customers, palette="Set2")

# Add lebals to the bars

plt.title("Customer Count per Segment")
plt.xlabel("Segment")
plt.ylabel("No. of Customers")
plt.xticks(rotation=45) # rotates the x axis bar lebals
plt.show()
```

```
Numbers of Customers in each Segments :


             Segment  Customer_Count
0              Loyal             542
1  Potential Loyalist             248
2          Champions              84
3             Others              77
4               Risk              47
5               Lost               2
```



In [30]:
```
t_revenue=rfm.groupby("Segment")["Monetary"].sum().reset_index()

print(f"\n Revenue Contribution by each Segment :\n")
print(f"\n{segment_revenue}\n\n")

# pie chart
plt.figure(figsize=(8,8))
plt.pie(segment_revenue["Monetary"],labels=segment_revenue["Segment"],autopct="%1.1f%%",colors=sns.color_palette("Set2"))
plt.title("Revenue contribution by segments")
plt.show()
```

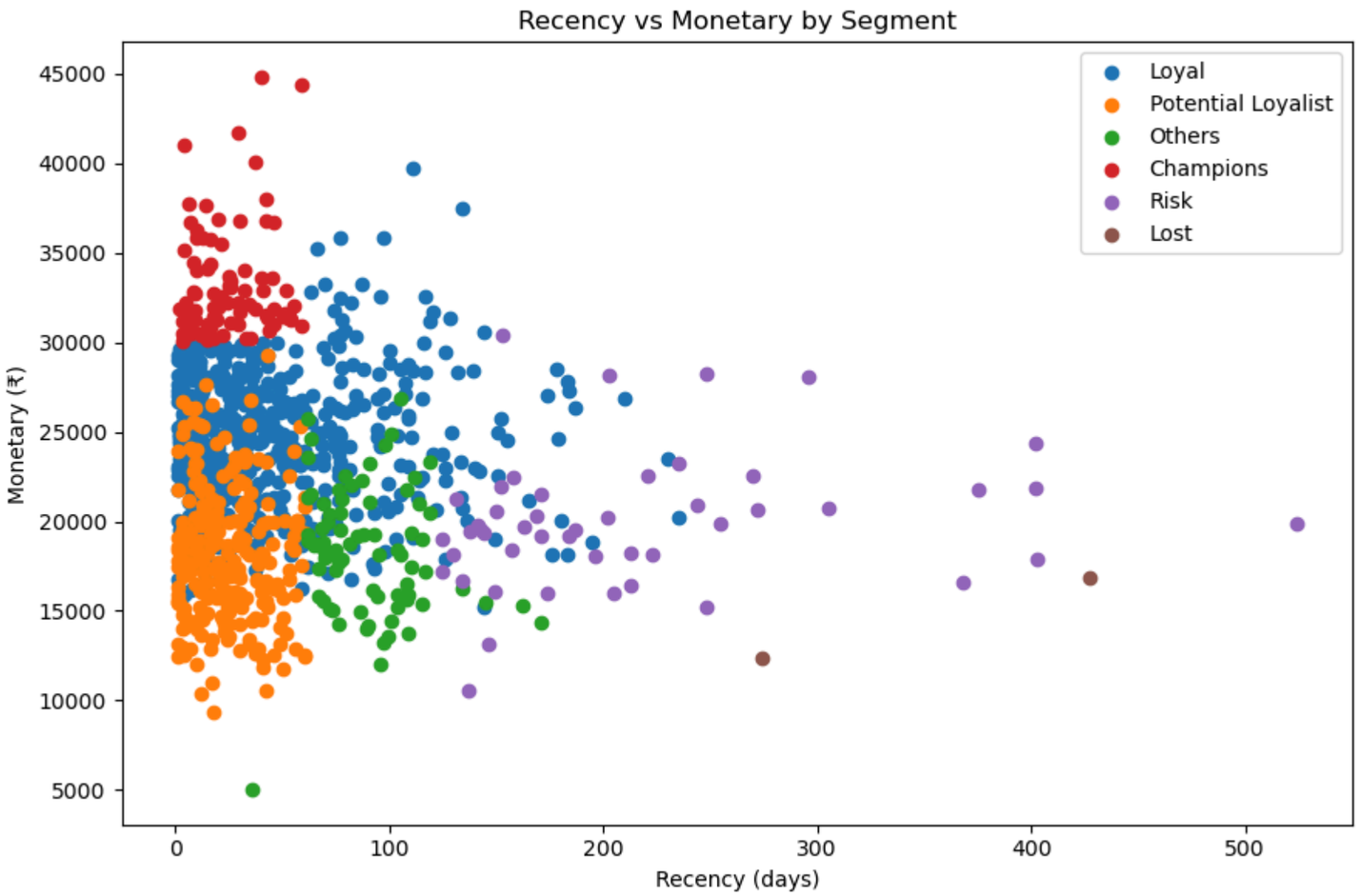
```
 Revenue Contribution by each Segment :


              Segment     Monetary
0           Champions   2791699.03
1                Lost     29221.39
2               Loyal  13381926.89
3              Others   1416376.98
4  Potential Loyalist   4495435.99
5                Risk    938539.38
```



In [31]:
```python
scatterplt=rfm[["Recency","Monetary","Segment"]]
print("\n\nRecency vs Monetary by Segment:\n\n")
print(scatterplt.head())

# figure size
plt.figure(figsize=(9,6))

for seg in rfm['Segment'].unique():  #Loop through each customer segment
    data = rfm[rfm['Segment'] == seg]  #Select only the rows for that segment
    plt.scatter(data['Recency'], data['Monetary'], label=seg)  #scatter points

#Add labels and title
plt.xlabel('Recency (days)')
plt.ylabel('Monetary (₹)')
```

```
plt.title('Recency vs Monetary by Segment')


plt.legend()
plt.tight_layout()
plt.show()
```

```
Recency vs Monetary by Segment:


   Recency  Monetary             Segment
0       13  21265.49               Loyal
1       35  28654.31               Loyal
2       18  23884.03               Loyal
3       81  24206.03               Loyal
4        8  25565.30  Potential Loyalist
```

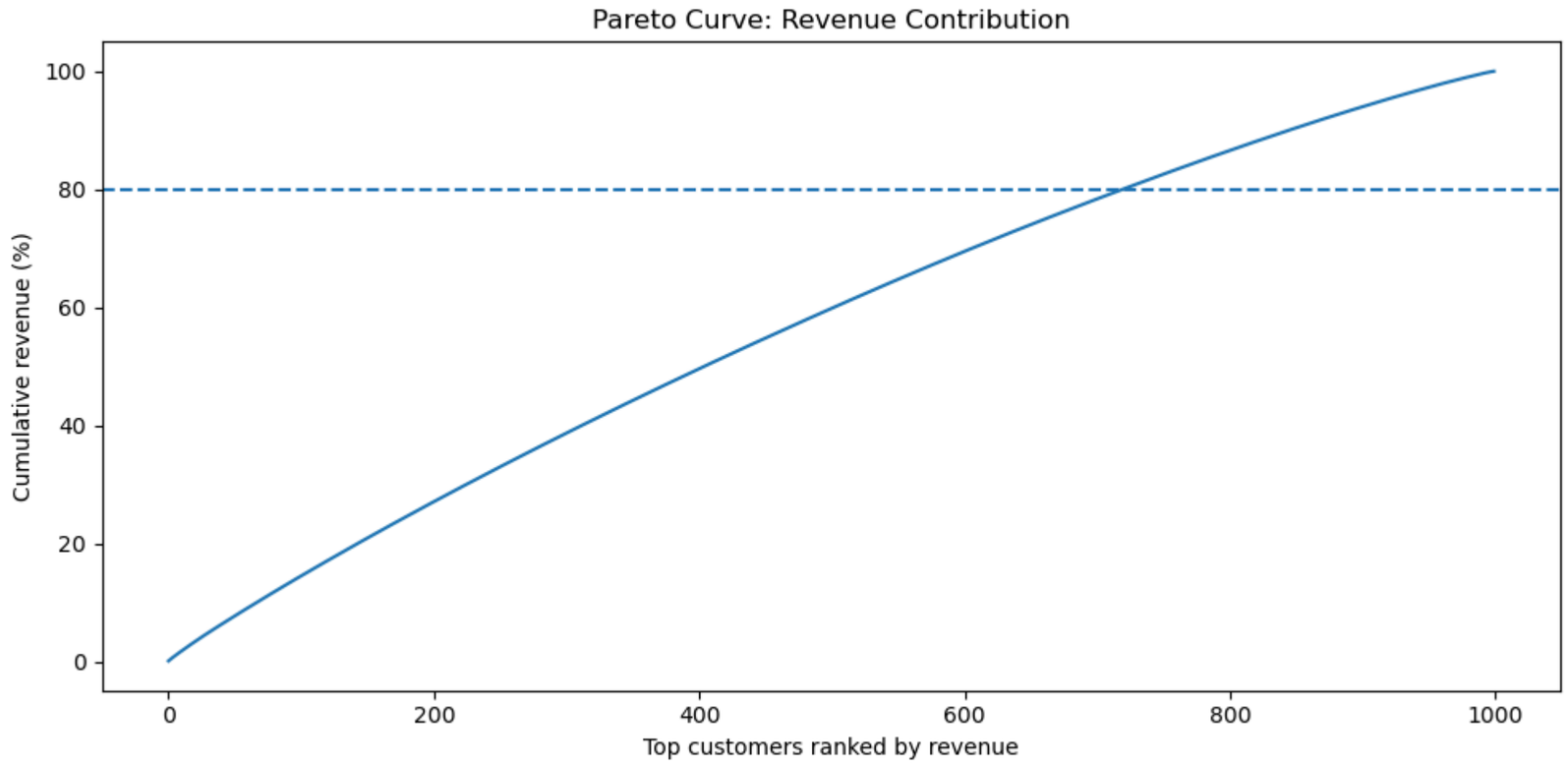


Pareto: Top 20% customers contribution

In [32]:
```
# Sort customers by revenue (highest first)
cust = rfm[['CustomerID', 'Monetary']].sort_values('Monetary', ascending=False).reset_index(drop=True)

# Calculate cumulative revenue
cust['cum_rev'] = cust['Monetary'].cumsum()

# Finding the total revenue
total = cust['Monetary'].sum()

# Convert cumulative revenue into percent
```

```python
cust['cum_pct'] = (cust['cum_rev'] / total) * 100

# Identifying top 20% customers by count
top20 = int(0.2 * len(cust))           #0.2 * len(cust)represent If 100 customers → top 20 customers
                                        #If 500 customers → top 100 customers means total len multiplied by .2 to get top 20

# Calculate their revenue contribution
top20_pct = cust['Monetary'][:top20].sum() / total * 100 # what % of total money came from top 20% of customers

print(f"\n\nTop 20% customers contribute {top20_pct:.2f}% of revenue\n\n")

# Pareto curve

plt.figure(figsize=(10,5))
plt.plot(cust['cum_pct'])  #line plot

plt.axhline(80, linestyle='--')  #horizontal dashed line at 80%

plt.xlabel("Top customers ranked by revenue") #Label for x-axis
plt.ylabel("Cumulative revenue (%)")          #Label for y-axis
plt.title("Pareto Curve: Revenue Contribution")  #Adds a title on top of the graph.
plt.tight_layout() #Adjusts spacing (labels and title fit neatly)
plt.show() # isplays curve

print("\n\n\n==========================================================THANKYOU==========================================================================")
```

```
Top 20% customers contribute 27.03% of revenue
```




```
==========================================================THANKYOU==========================================================================
```

In [229…

```python
#                           >>>>>        Key business insights & recommended actions       <<<<<
```

```
#  1. Champions & Big Spenders:

#                            Only 27% of revenue comes from the top 20% customers (not 80% like many businesses)
#                            Revenue is spread out, so you should not focus only on top customers.
#                            Mid-level customers also matter and should receive offers or deals.

#  2. Recent Buyers vs Hibernating Customers:

#               Recent Buyers:
#                             Bought recently but not many times.

#                            > Good chance to turn them into loyal customers
#                              with welcome offers or product suggestions.

#               Hibernating Customers:
#                                     Haven't bought for a long time and buy rarely.

#                                   > Send light reminders or small offers—don’t
#                                     spend too much marketing budget here.

#  3. Potential Loyalists:

#                These customers:

#                                 > Buy moderately often

#                                 > Buy somewhat recently

#                                 > Spend a fair amount

#            >> They are very close to becoming Loyal Customers <<

#    Give them reward points, free shipping, or personalized product suggestions.



# 4. At-Risk Customers:

#                     These customers used to buy often, but now they have stopped buying.

#                     Send them:

#                               > “We miss you” emails

#                               > Special comeback discount

#                               > Personalized messages

#                     Goal: bring them back before they churn.



# 5. Pareto & LTV:

#               In many businesses : Top 20% customers usually bring 80% of revenue.

#               But in data : Top 20% bring only 27% of revenue.

#               Result:
#                       revenue comes from many different customers.

#                       You should calculate Customer Lifetime Value (LTV) for each segment.

#                       Also perform cohort analysis to see how customers behave over time.
```

In [ ]: